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KANSEI ENGINEERING AND WEB SITE DESIGN

Zheng Song
Technical University of Denmark
Lyngby, Denmark

Thomas J. Howard
Technical University of Denmark
Lyngby, Denmark

Sofiane Achiche
École Polytechnique de Montréal Montréal,
Québec, Canada

Ali G. Özkil
Technical University of Denmark
Lyngby, Denmark

ABSTRACT
Capturing users' needs is critical in web site design. However, a lot of attention has been paid to enhance the functionality and usability, whereas much less consideration has been given to satisfy the emotional needs of users, which is also important to a successful design. This paper explores a methodology based on Kansei Engineering, which was significant used in product and industrial design but not quite been adopted in the IT field, in order to discover implicit emotional needs of users toward web site and transform them into design details. Survey, interview techniques and statistical methods were performed in this paper. A prototype web site was developed based on the Kansei study results integrated with technical expertise and practical considerations. The results showed that the Kansei Engineering methodology, in this paper, played a significant role in web site design in terms of satisfying the emotional needs of users.

Key words: emotional design, Kansei Engineering, web site design.

1 INTRODUCTION
The Internet has become indispensable in modern life, with a proliferation of different web sites flooding our screens. Web site design is a very complex process usually involving multiple technologies and knowledge disciplines. One of the important skills of a web site designer is to produce functional sites that also evoke the desired emotion in users. Theoretically, users' satisfaction and technical factors are both important to a successful design (Nagamachi, 1992). Therefore, to understand and satisfy people's needs is one of the fundamental aspects in design.
According to Jordan, there exists a hierarchy of needs explaining what kinds of expectations products should meet (Jordan, 2000). Three levels of needs are included in the model as in functionality, usability and pleasure as illustrated in Fig. 1.

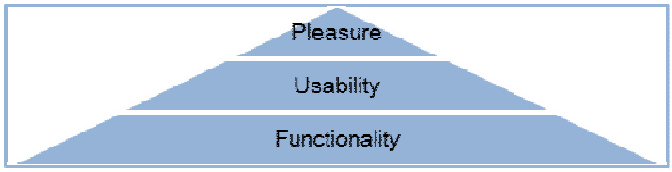

Figure 1 JORDAN'S HIERARCHY OF NEEDS (Jordan, 2000)

With the lower level fulfilled the request for higher level comes to users. The most primary need from a product is to function (Howard et al 2008, 2010, 2011), which is the core basics of a product, depending on the quality; with the need of functionality satisfied, people want the product to be easy to use (usability). The third level, pleasure, refers to emotions that products bring to people. As Jordan said, pleasure-based approaches are about understanding and respecting what people want from products and designing something that can "bring a real joy into people's lives." (Jordan, 2000) Researches have shown that emotions aroused from products influence the pleasure of buying, owning, and using them (Hirschman & Holbrook, 1982). Furthermore, in products one could link elements of the design to specific reactions of the users (Achiche & Ahmed, 2008, 2009, 2010), this means one can dine tune design features to make sure the user gets the intended emotional experience.
Web sites that bring positive emotions to users must meet all three levels proposed by Jordan. With the evolution of technology, functionality and usability can be largely improved and perfected; however, compared to the first two levels, the level of pleasure is much harder to achieve. Although increasing attention is being paid to the emotional aspects of web site

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design, there is currently not a proven systematic method of determining design concepts and details (Lokman, 2011).
Lokman has made an attempt to engineer emotion in web site design using Kansei Engineering, which has done significant work in product and industrial design but not quite been adopted in the IT field. The research presented a structure of Kansei Engineering in the context of web site design and provided an example of designing online clothing web sites (Lokman, 2011).

1.1 What Is Kansei Engineering?

The expression "Kansei" has its origins in Japanese with no direct translation in English. Nagamachi explains Kansei as "an individual's subjective impression from a certain artefact, environment or situation using all the senses of sight, hearing, feeling, smell, taste as well as recognition" (Nagamachi, 2001).
As the name suggests, Kansei Engineering regards technology that integrates Kansei with the engineering realms to penetrate people's Kansei into product design so that the products can trigger emotional responses and satisfy users (Lokman et al, 2009). The appeal of Kansei Engineering is that it presents a systematic way of collecting and understanding people's feelings and perspectives toward certain products through a series of physiological and psychological measurement and scientific methods (Lokman, 2010).

1.2 Kansei Measurement

Kansei, which is influenced by people's experience, knowledge, personality and sentiment and differs from person to person, is too implicit to be measured directly. Some standard measurement methods were developed based on externalisation, and have been successfully adopted in specified areas.
Methods that stand out among all the alternatives in terms of converting intangible Kansei into product development information are preferably based on semantic descriptors (Schütte, 2005), where Semantic Differential Scale by Osgood (Osgood et al, 1957) is one of the most widely used measurement methods partly due to its versatility (Himmelfarb, 1993). Even though this method is not objective enough, it is more suitable than physiological measurements for the purpose of identifying a rough picture of Kansei (Nagasawa, 2002).

1.3 General Model of Kansei Engineering

Kansei Engineering has been well adapted in industrial design in Japan and Korea (Lokman et al, 2009). Even though different types of procedure are applied in various contexts, a general model of Kansei Engineering methodology was developed by Schütte (Schütte, 2005) based on the examinations of different Kansei Engineering applications.
In this model, the chosen product domain is covered with both a semantic and physical perspective, in the form of 'Semantic Space' and 'Space of Properties'. Semantic Space is a vector space defined by certain words, where products can be described. The words are selected by manual or mathematical methods. The Space of Properties is a collection of products representing the domain, and key features and properties of these products will be identified for further evaluation. By implementing different techniques into the steps, Kansei can be linked with product properties. After the test of validity a model will be built.

1.4 Objective of the Research

Seeing the importance and potential of embedding emotions into web site design and the possibility of using Kansei Engineering as a requirement generation technique to improve user experience, this paper is trying to explore a Kansei Engineering methodology that can transform implicit emotions and needs of users into design schemes to support web site design in terms of achieving the level of "pleasure".
Inspired by Lokman's research paper Kansei/Affective Engineering and Web Design (Lokman, 2011), this paper employs its methodology as a blueprint but implements the steps with different approaches and analyse data using statistical methods from different angles. In addition, Lokman's paper lacks an explanation of advantages and limitations of using certain kinds of methods and technologies, which will be supplied in this paper. Additionally, with further studies and experiments, the methodology used in the paper can be applied in other domains to help designers better understand users' desires and design the products users really like.

2 RESEARCH METHOD

To help understand the research method, explanation of key terms used in this paper is given below (see Fig. 2):

1. Kansei words: words (usually adjectives) selected from people describing their Kansei toward a certain product.
2. Semantic Differential scale: a rating scale between two bipolar adjectives with 5 or 7 rating points.
3. Kansei checklist: a Kansei measurement tool developed by organising Kansei words into semantic differential scales.
4. Semantic Space: a vector space where products can be described, defined by certain words. The certain words are selected by manual or mathematical methods.
5. Design element: the parts and units within the basic structure of a web site.
6. Items: the properties of design elements.
7. Categories: specific attributes of items in a web site.

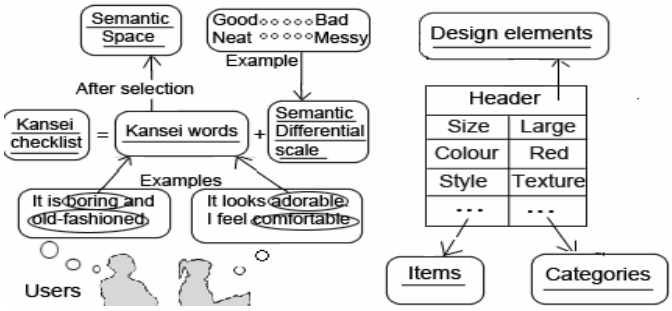

Figure 2 KEY TERMS USED IN THE PAPER

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The main steps of the research method are explained here, with each step discussed in detail in the following sections.

(1) Synthesising specimens. This part started with the task of collecting samples with visible differences from existing products in the market, following by investigation of design elements in all samples.
(2) Establishment of Kansei checklist. The Kansei checklist was developed as an instrument used in Kansei measurement procedure.
(3) Kansei measurement. This step was performed to discover people's Kansei responses when interacting with different web sites.
(4) Design scheme analysis. The result of Kansei measurement was used to find links between Kansei and web site design after being analysed using mathematical methods.
(5) Prototyping and testing. Based on the results from last step, a web site prototype was built and tested.

This paper chose to improve the appearance of the web site of Engineering Management department of the Technical University of Denmark (DTU) as a sample project to experiment the implementation and effects of applying Kansei Engineering to web design. The project focused on the homepage of the web site because it gives a general impression of the whole site and other pages can easily follow its style.

3 SYNTHESISING SPECIMENS

Web site specimens were selected and synthesised for two purposes. First, properties and key features of web sites were identified by investigation of design elements. Second, the specimens would be used for the evaluation session. The procedure involved two parts, collection of specimens and investigating the design elements.

3.1 Collection of Specimens

The specimen selecting procedure was based on random collection but careful considerations, hence, criteria were identified as follows:

1. Since this paper was working on the web site of engineering management department, specimen selection should focus on the similar domain, which should be web sites of business schools or management departments of universities.
2. There should be visible differences between selected web site specimens, in terms of colour, layout, typography, etc.
3. The language used by the specimens should be English.
4. Brand effect should be decreased as much as possible. People may have intention to like a web site not because of its good design but the university's fame. Therefore, well-known universities should not be considered in the collection.
5. Due to limited time and other resources, number of specimens cannot be too large.
6. The web site of engineering management department of DTU is included as a specimen. Evaluation will be given to each specimen from later procedures and it will be helpful to know what users think about the current design of the web site.

On the basis of these criteria, nine American universities ranked from 300 to 500 on the list of "Ranking Web of World Universities" (http://www.webometrics.info/) were randomly chosen. Reasons behind the decision were that: first, universities ranked from 300 to 500 are not well-know enough but still count as good ones; second, it would be easier to find non-American test subjects around DTU, who probably had not heard of these less popular universities from USA. Together with the web site of DTU management, ten specimens were selected, waiting for further investigation. A snapshot of the specimens is shown in Fig. 3.

Figure 3 SNAPSHOT OF SPECIMENS

3.2 Investigation of Design Elements

In the context of web site design, design elements mean the parts and units within the basic structure of a web site that are commonly used by web designers and understood by web site users, like header, menu or main body.

To get a clear categorisation of the design of web sites, these elements were broken down into items and categories, summarised from all of the specimens. As the illustration in Fig. 1 indicates, items regard the properties of each design element such as header size and header background colour, and categories mean the specific attributes of items in each specimen, for example, the header background colour of a specimen is red, yellow or blue.

As shown in Tab. 1, 30 items were identified according to the characteristics of the specimens. Each specimen was investigated based on the 30 items to form the matrix of category. Table 2 demonstrates the categories of the ‘header background colour’ item, in which the mark ‘X’ means the category fits the specimen.

With identification of items and categories, key features and properties of the specimens were well organised and recorded, and the categories would be used as independent variables for statistical analysis, which could contribute to discovering the relations between Kansei and web sites and proposing a design scheme.

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Table 1 ITEMS IDENTIFIED FROM SPECIMENS

No	Header background colour							
	red	dark blue	blue	black	dark green	green	white	black& red
1	x							
2		x						
3			x					
4				x				
5					x			
6						x		
7							x	
8								x
9			x					
10							x	

Table 2 EXAMPLE OF CATEGORIES

Design elements	Specification	Items
Page	Page of a web site	Page background colour / size
Header	Head section of a web site	Header background colour / background style/ size
Top menu	Menu on top of a web site	Top menu existence / background colour / font size / font colour / style
Left menu	Menu on the left part of a web site	Left menu existence / background colour / font size / colour / style
Main body	Main section of a web site	Main body background colour / title font colour / title font size / text font colour / text font size
Footer	Bottom section of a web site	Footer size / background colour / font colour / font size
Picture	Pictures displaying of campus life	Number of Picture / number of large picture / number of medium picture / number of active picture
Title	College title	college title font size / college title font colour

4 ESTABLISHMENT OF KANSEI CHECKLIST

The Kansei checklist is a very important tool in Kansei Engineering that can measure intangible Kansei by manageable scales with words. The focal point in this step is the mechanism of synthesising Kansei words.

4.1 Synthesising Kansei Words

The process of synthesising Kansei words is rather complicated since the words can not only describe product domain (Schütte, 2005) but also imply users' emotional responses towards a certain product. Thus, more attention should be paid on gathering the adequate Kansei words for our case study.

This section took a more rigorous and systematic approach for choosing the proper Kansei words than the previous research by Lokman which the paper builds on. In Lokman's paper (Lokman, 2011) a straightforward approach for the selection of words based on frequency of appearance in web design guidebooks, web sites, research papers and journals, and general words related to describing web sites. A common technique used in Kansei Engineering is to collect words from various sources until no new words come out and then span the structure of Kansei words manually by an expert or statistical methods (Schütte, 2005).

The approach used in this paper for synthesising Kansei words contained three steps and two filters. For the purpose of implementation of this approach, a literature review on Kansei Engineering and web design was conducted and methods such as interviews and Kappa test were also used. From Fig. 4 the structure of this approach can be clearly grasped. Details of implementation will be described next.

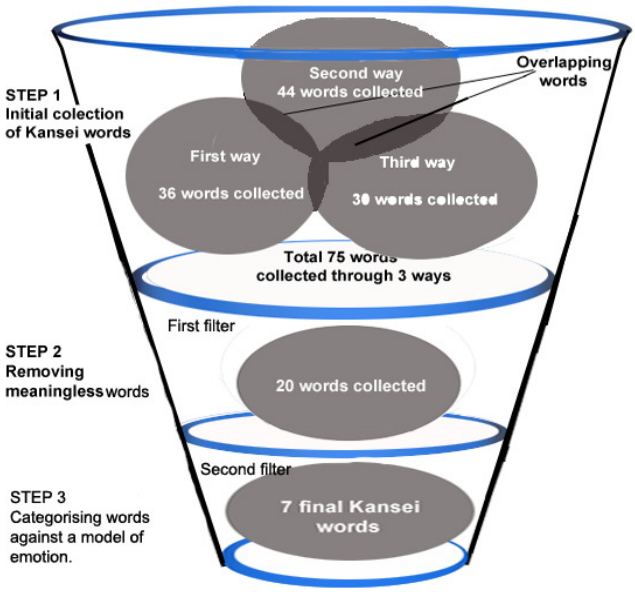

Figure 4 MODEL OF SYNTHESISING KANSEI WORDS

STEP 1: Initial Collection of Kansei Words. As indicated in Fig. 4, a set of Kansei words were pulled together through three different ways:

1. Words collected from articles regarding principles of web design and research papers on emotional design.
2. Students from the department of management (who are the main users of the DTU management web site) and computer science (who have knowledge and experience of web design) were interviewed and asked to write down words that describe their feelings and expectations of web sites based on their experience of surfing the internet.
3. Five web sites of universities below 500 on the list of "Ranking Web of World Universities" were randomly chosen as testing samples. Participants were asked to view these samples for a few minutes and then describe their feelings about these web sites using adjectives.

A total of 75 words were gathered including both positive words (e.g. pretty) and negative words (e.g. boring). The more times a word was mentioned by the participants or articles, the stronger power it possessed.

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STEP 2: Removing Meaningless Words (first filter). This filter aims to remove words that do not fit in the context of web design and that have similar meaning. By doing this words passing through the filter will be more significant and suitable for describing the emotions evoked by web sites.

First, ten words which were not suitable for describing appearance of web sites were removed such as 'fast' and 'visual'. Second, by using the Oxford dictionary and "WordNet"-a database for English language that groups words into sets of synonyms (http://wordnet.princeton.edu/), the rest of the words were sorted by similarity and dissimilarity of meaning, i.e. synonym and antonym, which ended up with 20 different groups. Figure 5 is an example of word grouping. Words in brackets are antonyms of others in the same group.

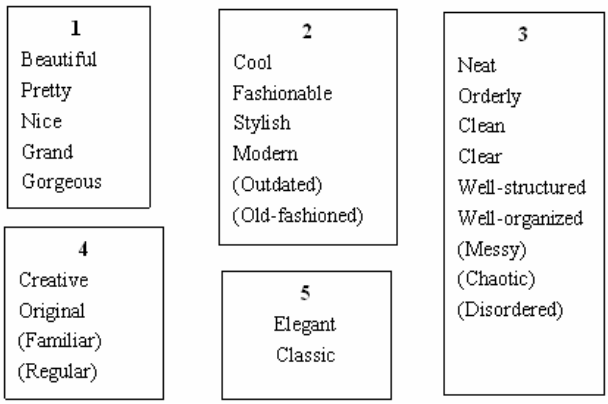

Figure 5 EXAMPLE OF WORD GROUPING

On the basis of aforementioned word categorisation, twenty words were picked out according to the following rules:

1. From each group the word with strongest power is selected. If more than one words have the same power, one is randomly picked.
2. Words that do not have synonyms and antonyms in the initial collection are all selected.
3. Only choose positive words as they will later be paired to their antonym.

STEP 3: Categorising Words Against A Model of Emotion (second filter). The idea of the second filter is to categorise the twenty selected words against a model of emotion, by which typical reactions of users to products are categorised based on theoretical thoughts. Thus, words taken from each of the categories will be more representative and comprehensive.

Emotions have been classified by different scientists at different times in a number of ways and theories. After careful consideration and comparison of many models, Goldman's (Goldman, 1995) eight categories of evaluative aesthetic terms were chosen because they are used for evaluating aesthetic value in general and art criticism and evaluation in particular, which is appropriate in web design since it is a very visual product. Moreover, the categories are helpful for expressing the essence of what aesthetic terms convey. (Pham, 1999) The eight categories with examples are summarised in Tab. 3.

Table 3 GOLDMAN'S CATEGORIES (Goldman, 1995)

Category	Example
Broadly evaluative	beautiful, ugly, sublime, dreary
Formal	balanced, graceful, concise
Emotional	sad, angry, joyful, serene
Evocative	powerful, stirring, amusing, hilarious, boring
Behavioural	sluggish, bouncy, jaunty
Representational	realistic, distorted, artificial
Perceptual	vivid, dull, flashy
Historical	derivative, original, conservative

However, the categories of 'Representational' and 'Historical' did not fit the context of describing emotions evoked by web site and for that reason only the other six categories were used.

To assess the quality of the classification of the words into the six categories a Cohen's kappa test was used. Two master students from DTU management were invited to join the experiment. Their task was to classify the twenty words into the six categories based on their comprehension and experience. Each word could only be seen in one category and participants were allowed to skip words that they did not know or could not decide where to put.

The kappa value was calculated at 0.43, which, according to Altman (Altman, 1991) means that the strength of agreement between the participants was moderate. Thus the categorisation result was valid. Words that were put in same categories by both participants were picked out and grouped into the six categories mentioned above. The final Kansei words in Tab. 4 were selected by the following rules:

1. Choose the word with strongest power from each category.
2. Two words are selected from the category "Broadly evaluative" because this category is very general and the two words have equally strong power.
3. Make the words into bipolar pairs for the purpose of developing Kansei checklist later.

4.2 Development of Kansei Checklist

Seven pairs of Kansei words were organised into 5-point semantic differential (SD) scales in order to form the Kansei checklist. The scale values were 2, 1, 0, -1, and -2 so that evaluation responses would be conveyed to these values. But test subjects would not be informed about the scale values. The structure of the Kansei checklist can be seen from Tab. 5.

Table 4 FINAL KANSEI WORDS

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Kansei Words		Category
Beautiful	Ugly	Broadly evaluative
Modern	Outdated	Broadly evaluative
Well-structured	Disordered	Formal
Enjoyable	Unpleasant	Emotional
Exciting	Boring	Evocative
Dynamic	Lifeless	Behavioural
Bright	Dull	Perceptual

Table 5 KANSEI CHECKLIST

	2	1	0	-1	-2	
Beautiful	○	○	○	○	○	Ugly
Modern	○	○	○	○	○	Outdated
Well-structured	○	○	○	○	○	Disordered
Enjoyable	○	○	○	○	○	Unpleasant
Exciting	○	○	○	○	○	Boring
Dynamic	○	○	○	○	○	Lifeless
Bright	○	○	○	○	○	Dull

5 KANSEI MEASUREMENT

This section was conducted by an online survey consisting of the Kansei checklist as the measurement tool and the specimens with test subjects.

5.1 Online Survey Design

The idea of the survey was to show links to each of the specimens with ten sets of Kansei checklist. Test subjects would be asked to click every link and view the web site carefully for several minutes and then rate their feelings toward the web site on the checklist. The same activity continued until all specimens were viewed and evaluated.

Though face to face interview would be the best way to do the survey, after considering time and resource limitations an online survey was thought to be the better choice as a quicker and easier way to recruit more subjects in a short time.

To minimise bias and inertial thinking during the process, the order of Kansei words on the checklist would be changed for every web site. (Lokman, 2011) Since the examination was about the appearance of the homepage, there was no need to touch any of the menus or links on the web sites.

A session of pilot test was carried out with two professors and one master student at DTU to validate the survey. Based on the feedback and suggestions obtained from the test, the survey was improved to have a picture show including snapshots of the ten specimens before questions being answered, to let test subjects have an overview of what the web sites look like.

5.2 Web Site Evaluation Session

The survey was sent out through email, SMS and social networks. 45 test subjects answered the survey within nearly one month, among which 36 subjects finished all the questions while the other 9 left part of the survey unanswered.

5.3 Detecting Outliers

Grubb's test is a statistical method to detect outliers in a univariant data set assumed coming from a normally distributed population, which spots outliers at one round at a time. To run this test, two issues had to be clarified. First, data gathered from the survey was assumed to obey normal distribution because Grubb's test is based on the assumption of normal distribution. Second, the test would be processed seven times since the Kansei checklist contains seven pairs of words but Grubb's test can only deal with a univariant data set.

The test resulted in four responses being removed of the data set and the remaining 32 sets of data would be used for design scheme analysis. The test subjects were aged between 18 and 30 years old with an equal distribution of gender. In Kansei Engineering methodology the population of subjects varies depending on objectives and measurement tools. (Lokman, 2011) It was suggested that the number of subjects for this kind of research was around 30 - 50. (Nagamachi & Lokman, 2010) Therefore, the result of having 32 subjects was acceptable for this survey.

6 DESIGN SCHEME ANALYSIS

The aim of this step is to describe the integration of Kansei and web site design, and analyse the design scheme for improving the web site of engineering management department of DTU. Multivariate analysis was used to uncover the relationships between Kansei and web sites, whose outcomes provided guidelines for new design. Factor Analysis was used to propose a general outline while Partial Least Square analysis was performed to present detailed guidelines. These two methods were also used in Lokman's research paper (Lokman, 2011). However, in this paper we investigated the analysis results in a different way, and the explanation of advantages and limitations of carrying out the analysis was also provided in details.

For the sake of convenience, we use positive words from each pair of Kansei words to represent the Kansei checklist. For example, "beautiful" means the scale of "beautiful ----- ugly".

To do the multivariate analysis, the mean value of evaluation data obtained from the 32 survey responses was calculated for each Kansei word of each specimen. The decision of using mean value over median was based on thoughtful consideration discussed next.

A Semantic Differential (SD) scale was hoped to be treated as an ordinal scale from statistical angle. However, the ordinal measurements only describe order, not degree of difference between measured items, so there are arguments about using mean or median value when operating on ordinal scales. On the one hand, median value, the middle-ranked item, are commonly used to describe the central tendency of a group of items when using ordinal scales, but precision and dispersion of the data will be affected since values of other items are not involved at all. Mean value, on the other hand, does not have a valid interpretation on ordinal scales, and it is often influenced by outliers that may be at the extremes of the data set. But mean

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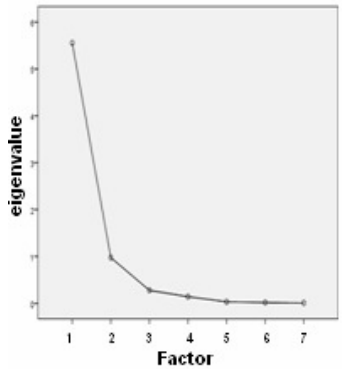
value is also more informative than median value because it uses all the data to do the calculation.
However, it is also argued that SD scale can be seen as interval scale because the neutral response serves as an arbitrary zero point and the intervals between scale values can be treated as equal. Thus, it was acceptable to use the mean value, which was valid for interval data. Besides, outliers have already been screened by Grubb's test in the last step. Therefore, considering the analysis upwards, mean value is used to analyse data by statistical methods in this step.

6.1 General Design Outline

This section was performed by Factor Analysis and was divided into two parts. First, the Semantic Space, a vector space defined by certain words where web sites can be described, was spanned. And then, within the Semantic Space the specimens were evaluated and ranked, and the highly rated specimens could arouse stronger positive Kansei responses than others. Thus, characteristics of these specimens were summarised to generate a general design outline.

Spanning the Semantic Space. Despite great endeavour to synthesise the Kansei words, they still could not be used for the Semantic Space directly because it is believed that redundancy of words, i.e. some of the words are correlated with one another, will happen due to the subjective perception. (Lokman, 2010) Therefore, methods that can cut down redundant information by synthesising words was required, which would transfer Kansei words that may have relations to each other into much fewer unrelated factors that could make the Semantic Space easier to interpret.
Factor Analysis (FA) is a statistical method widely used in social science, product management and other applied science. The basic idea of FA is to discover relations between observed, correlated variables and then simplify them with lower number of latent, uncorrelated variables, i.e. factors.
SPSS 20 (statistical analysis software) was used to undertake the FA and Principle Component Analysis was chosen as the extraction method. The variables were the seven Kansei words, and the data input was the mean value of the 32 survey responses for each Kansei word of each specimen.
The screen plot in Fig. 6 (left) shows the eigenvalue of each factor plotted from largest to smallest. When working on a screen plot, we look for a "bend" in the eigenvalues to decide where to stop the retention of factors. Thus, for the data above, it was clear that the first two factors should be retained.
The result of factor loadings shown in Fig. 6 (right) indicates the correlation coefficients between Kansei words and the two retained factors, where a high score means a word is highly related to a factor. Therefore, seven Kansei words were classified into two groups: the group of factor 1 consisted of *dynamic, exciting*, *bright* and *modern,* and the group of factor 2 included *beautiful*, *enjoyable* and *well-structured.* Hence, the Semantic Space was spanned, using the two factors as axes (See Fig. 7).



Kansei Words	Factor	
	1	2
dynamic	.964	.146
exciting	.807	.420
bright	.831	.490
modern	.712	.440
beautiful	.259	.880
enjoyable	.435	.836
well-structured	.167	.768

Figure 6 SCREEN PLOT OF FA (LEFT) AND FACTOR LOADINGS (RIGHT)

How the ten specimens spread in the Semantic Space is shown in the scatter plot (Fig. 7) according to their factor scores. In the plot it is clear that the specimens can be divided into four groups. Web sites at the right top corner (group 1) have higher scores on both of the criteria so they can be reckoned to evoke the strongest positive Kansei responses among the 10 web sites. On the country, web sites at the left bottom corner (group 3) are not so good at arousing positive Kansei responses. The other two groups, group 2 and group 4 are only considered good with respect to one criterion, so their ranking is higher than group 3 but lower than group 1.

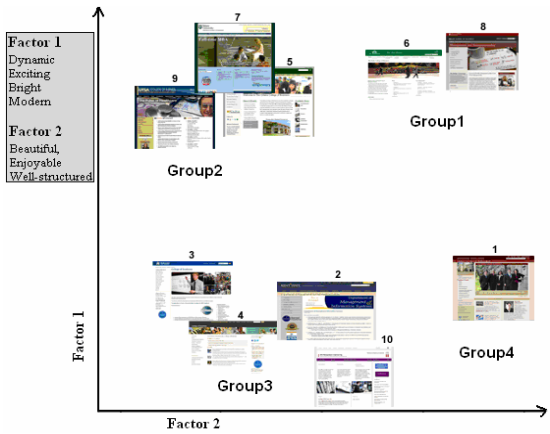


Figure 7 SCATTER PLOT

Proposing A General Design Outline. The result of FA offered an overview of relations between Kansei and web site designs and a basic structure of design scheme. We could identify characteristics from highly rated specimens, for instance, having more pictures especially large ones and distinct colour themes. Figure 8 is a general outline for the new design summarised from specimens with high marks. However, these were not enough to provide specific guidelines, so further analysis to uncover associations between Kansei words and detailed design in terms of items and categories would be performed by Partial Least Squares analysis.

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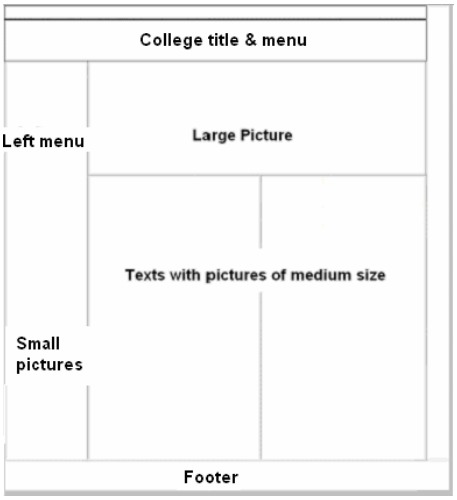

Figure 8 GENERAL OUTLINE FOR NEW DESIGN

6.2 Detailed Design Guideline

Partial Least Squares (PLS) analysis is a statistical method used to find relations between two matrices by creating a linear regression model. Comparing to other regression methods, PLS analysis can provide more rational and informative results, especially when the input data set is not big and the variables are related (as shown in FA some Kansei words are related to each other). Therefore, in this case PLS was applied to link the Kansei words with design items and categories so that we could find out guidelines for the new design.

PLS analysis was also conducted by SPSS 20. The dependent variables were the seven Kansei words, whose values were the same as used in FA analysis. The independent variables were the categories concluded from section 3. For the purpose of using PLS, the values of categories were dummy variable, i.e. 1 or 0. For instance, a specimen had gray page background colour so its value of this category was 1; while another specimen that did not fit this category, had the value 0.

Calculation of PLS analysis was also conducted by SPSS 20. Table 6 (excluding the columns of 'mean value' and 'variance') demonstrates an example of the analysis result regarding 'header background colour', indicating how the categories influence people's responses on each of the seven Kansei words: higher score means bigger influence and minus numbers show negative impact.

To form a detailed guideline one category was chosen for each item. The way to choose categories was that: for each category calculate the mean value and variance of the parameters regarding the seven Kansei words, and then choose the categories with bigger mean values. If more than two categories had very comparable mean values, the one with smaller variance would be chosen. The reason behind this was that the category with bigger mean value had stronger positive impact on people's Kansei toward web sites, and smaller variance showed the impact on the seven Kansei words was more balanced. In this way we intended to design a web site that could present both a combination and a balance of all the seven Kansei words. As shown in Tab.6, the category 'dark red' was selected for the header colour. Detailed guidelines can be seen in Tab.7, where a specific category of each item is indicated.

Table 6 EXAMPLE OF PLS ANALYSIS RESULT

Categories	Kansei words (dependent variables)								
Head background colour	Modern	Well-structured	Enjoy-able	Excit-ing	Dyna-mic	Bright	Beauty-ful	Mean value	Variance (10^{-4})
Dark red	0.041	0.073	0.04	0.017	0.014	0.029	0.064	0.04	5.0
Dark blue	0.029	-0.002	0.008	0.023	0.042	0.018	0.008	0.018	2.2
Blue	0.029	0.023	0.044	0.051	0.022	0.025	0.05	0.035	1.7
Black	0.013	-0.033	-0.022	0.019	0.02	0.011	-0.041	-0.005	6.9
Dark green	-0.029	-0.024	-0.003	-0.026	-0.027	-0.02	-0.011	-0.02	0.92
Green	-0.066	-0.076	-0.062	-0.051	-0.042	-0.063	-0.07	-0.06	1.3
White	0.013	0.021	0.017	0.012	0.005	0.01	0.017	0.014	0.28
Black &Red	-0.072	-0.056	-0.066	-0.067	-0.049	-0.04	-0.08	-0.06	1.9

Table 7 DETAILED GUIDELINE

Page background colour	Gray
Page size	Medium
Header size	Medium
Header background colour	Dark red
Header background style	Mixed with picture
College title font size	Medium
College title font colour	White
Top menu existence	No
Left menu existence	Yes
Left menu background colour	White
Left menu font size	Large
Left menu font colour	Black
Left menu font style	Bold
Main body background colour	Blue
Main body title font colour	Blue
Main body title font size	Medium
Main body text font colour	Black
Main body text font size	Normal
Footer size	Large
Footer background colour	White
Footer font size	Mixture of large and small font
Footer font colour	Blue & gray
Number of picture	4~6
Number of large picture	1
Number of medium picture	5
Dynamic picture	Yes

However, the result was obtained only from statistical calculating, so it is inevitable to have irrational combination of elements and categories in the guideline, for example, it is almost impossible to see the texts with the colour of blue in a blue background. Besides, a web site with higher scores does not mean that all of its elements are properly designed. Furthermore, how to merge the university's culture into the design should also be taken into account. For all these reasons, the final design scheme would be an integration of both guidelines from FA and PLS and other considerations, and of course designers' creativity and inspiration also matter.

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7 PROTOTYPING AND TESTING

This step developed a web site prototype using the results from last step with basics of Graphic Design and tips of web site design as well as practical considerations. Then a test was played to validate the implementation of Kansei Engineering in web site design.

7.1 Prototyping

As discussed before, it is not rational to follow exactly what the guidelines state, so that the results from statistical analysis should only be used as suggestion and inspiration to designers. The new design also relies on technical expertise and practical considerations.

It should be noted that the text on the homepage of DTU management web site was retained in the prototype. Pictures were selected from the whole DTU management web site to reflect campus life at DTU management.

Figure 9 illustrates the prototype, whose basic structure was based on the general outline obtained from the FA result (see Fig. 8). Examples of how to combine and harmonize the guidelines with design expertise and practical issues are given.

1. In the detailed guidelines the colour of header was dark red, but it was not a proper choice in this case because part of the DTU logo (on the left side of the header) was also red. Therefore, blue, the second best choice (see table 8), was used as the header colour. Additionally, the application of colour gradient created a sense of depth, making the header seem not dull.
2. The search box and miniature picture at the right end of header played the role of keeping the header balanced.
3. Dynamic pictures (four large size pictures displaying in turn) were the focal point of the whole page, making the web site seem lively.
4. Five pictures of medium and small sizes were included to give a vivid impression to users.
5. The two units were given the same style to show their relations, and at the same time separate them from the left menu.

7.2 Testing

This section presents the testing of the success of the prototype, to see if there was any difference in users' Kansei responses after applying Kansei Engineering to web site design.

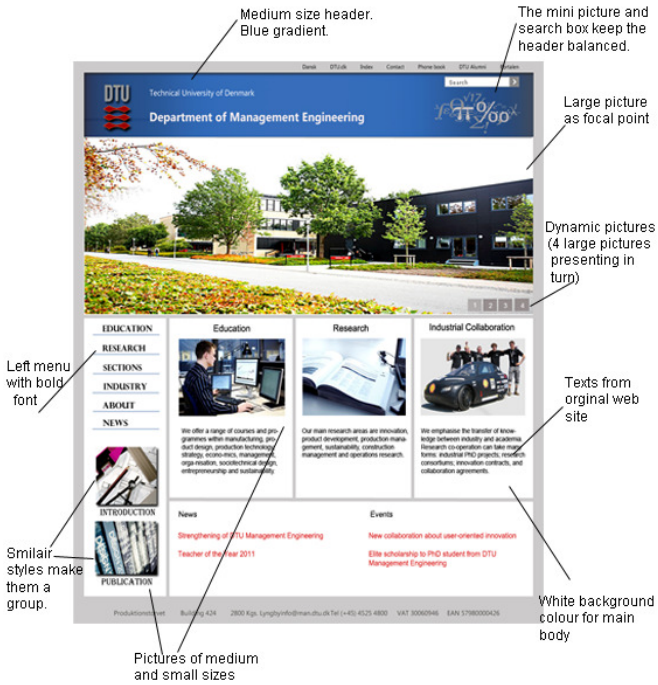

Figure 9 PROTOTYPE

Obtaining Evaluation Data. As it can be seen from the scatter plot in Fig. 7, ten specimens are divided into four groups, representing their different grades in terms of achieving the "level of pleasure". Hence, from each of the four groups one web site was chosen to complete the test with the prototype. The web site of DTU management was chosen from its group, so the test could not only indicate the ranking of the prototype among web sites with different grades but also check if the prototype has improved the original one and aroused stronger positive Kansei responses in users.

The five new specimens were evaluated following the same procedure described in section 5. In total 15 valid responses were selected from all of the answered surveys, and the data was analysed by FA again, whose result was used to make the comparative analysis. At this time the two factors describing the Semantic Space were *beautiful, bright, modern, enjoyable* (factor 1) and *dynamic, exciting, well-structured* (factor 2).

Comparative Analysis. From the new scatter plot (see Fig. 10), the prototype sits at the very top right corner, which means it aroused the strongest positive Kansei in users. Comparing to the web site of engineering management department of DTU (No. 10) which is at the bottom part of the plot, the prototype has made a significant improvement in terms of approaching the level of 'pleasure' to meet the users' needs.

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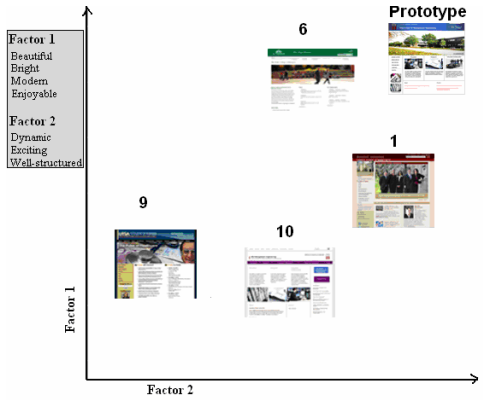

Figure 10 NEW SCATTER PLOT

The result has proved that the Kansei Engineering methodology used in this paper has successfully supported the appearance design of web sites, not only giving designers a better understanding of what users need but also guidelines and inspirations of designing web sites that evoke positive emotions in users.

8 CONCLUSION

The paper has explored the methodology built on Lokman's model (Lokman, 2011) and applied Kansei Engineering into web site design, by which users' needs and emotions were connected with the design details and the newly designed web site did evoke stronger positive emotions in users.

With the help of the Kansei checklist, a web site evaluation session was conducted to uncover the relations between emotional responses and web sites. Factor Analysis and Partial Least Square analysis were performed to generate guidelines for the design scheme. The result has proved that the Kansei Engineering methodology in this paper played a significant role in web site design in order to approach the "pleasure" level of users' needs.

8.1 Limitations

Due to multiple restrictions, the web site evaluation process was conducted by an online survey, which may suffer the following drawbacks: first, the target groups visiting the web site of DTU management are students, teachers and potential applicants of this department, but the online survey could not guarantee sources of test subjects. Second, the evaluation process was not able to be carried out in a controlled and systematic manner since it was largely depending on subjects' activities.

If conditions allowed, it would have been beneficial to have more specimens with different characteristics, which would offer more options for the new design.

Moreover, the case presented in this paper focused on an existing web site in trying to improve it, which represents more restrictions than creating a totally new web site.

8.2 Future Work

Further research and experiments will be performed in order to broaden the scope of application of the method used in the paper. Since web sites are also designed products, applying the methodology to other product design domain is very promising, only with changes of some details and methods to comply with the chosen product domain. Therefore, designers will have a better understanding of users' desires and can design the products users really like in a systematic and easier way.

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